Kunli Liu, PhD

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SUMMARY

AI-native staff engineer specializing in large-scale distributed infrastructure and production LLM systems. Builds end-to-end agentic AI platforms—from transformer-based agent design and inference serving to evaluation frameworks and AI-driven automation—with measurable impact on developer velocity and revenue. Leads cross-org engineering initiatives and drives technical direction from research to production.

Languages: Hack/PHP, C++, Python, PyTorch, SQL, PySpark
AI/ML: transformer architecture, attention mechanisms, BPE tokenization, pre-training, post-training alignment (SFT, RLHF, RLVR, DPO), scaling laws, CNNs, computer vision, object detection, feature engineering, distributed training (data/model/pipeline parallelism), GPU kernel optimization (CUDA)
AI Systems: LLM agent systems, agentic frameworks, tool-use orchestration, multi-step reasoning, model serving, inference optimization, prompt engineering, model evals, LLM observability & tracing, AI-driven automation
Data & Databases: SQL/PostgreSQL, NoSQL, Apache Spark, distributed data pipelines, ETL, query optimization, data modeling, Pandas, NumPy
Distributed Systems: orchestration, phased rollout, sharding, observability, reliability, E2E integration testing, CI/CD
Engineering Leadership: multi-team roadmap planning, cross-org partner alignment, headcount & capacity planning

EDUCATION

University of Pennsylvania — Philadelphia, PA
Master of Science in Computer Science
Selected Coursework: Computer Vision, Database & Information Systems, Big Data Analytics

Michigan State University — East Lansing, MI
Doctor of Philosophy

EXPERIENCE

Senior Software Engineer, *Meta Platforms* — 2022 – Present
Sunnyvale, CA
IC5 · Greatly Exceeds Expectations (top 10% performer)

Agentic AI Infrastructure for Ads Serving

- Architected and productionized a first-of-its-kind transformer-based autonomous agent with deep integration into the ads delivery stack, enabling autonomous reasoning over serving infrastructure telemetry at production inference scale.
- Built low-latency inference serving infrastructure for the agent—including context management, tool-use orchestration, and multi-step reasoning pipelines—and led cross-team adoption across pre-impression delivery services, establishing the foundational LLM agent framework for ads infrastructure teams.
- Validated agent efficacy through structured evaluation benchmarks measuring autonomous SEV detection accuracy and mitigation quality, projecting *$20M+* in annual revenue savings from reduced incident response time.

Unified E2E Test Platform — "TestPilot"

- Led 10+ engineers across ads infra, products, and change management to deliver a unified E2E testing framework spanning the full ad lifecycle: pre-delivery (publishing, indexing, matching, ranking) and post-delivery (impression, clicks, conversion, revenue).
- Owned TestPilot architecture; drove key design tradeoffs across deterministic vs. non-deterministic delivery and joint vs. isolated orchestration; designed LLM evaluation pipelines—scoring rubrics, benchmark suites, automated quality gates—to validate model-generated artifacts.
- Shipped **AI-TestPilot**, an agentic code-generation system using prompt-engineered LLMs for automated test authoring, validation, and enablement, driving ads targeting E2E test coverage from 0% to 95%.
- Launched TestPilot for pre-production ad lifecycle validation, directly addressing *$300M/year* in revenue exposure from reliability risks and product development inefficiencies.

Less Personalized Ads (LPA)

- Reliability area lead for LPA: built testing infrastructure from zero, where no prior reliability coverage existed, architected the full coverage model (canary, CI/CD integration, privacy API contracts).
- Drove launch-readiness decisions (gating criteria, failure modes, blast radius controls, exception handling), unblocking LPA launch and preventing *$10M* revenue loss.
- Shipped observability platform for LPA covering 100+ delivery-critical metrics across pre-commit, deployment, and production stages; defined SLOs, dashboards, and alert thresholds to reduce incident risk and shorten detection/triage loops.

Software Engineer Intern, *Oracle Corporation* — 05/2021 – 08/2021
Seattle, WA

- Built an event-driven E2E test automation pipeline for Oracle Hardware validation, triggering fleet-level validation on hardware events and eliminating manual test cycles for release qualification.